

FA-EEGNet: Enhancing EEGNet with Frequency-Adaptive Kernels for the classification of stroke rehabilitation movements

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Abstract—Brain-computer interfaces (BCIs) can enhance post-stroke rehabilitation by providing a direct communication pathway between the brain and assistive devices. Motor imagery (MI)-based BCIs typically rely on electroencephalography (EEG) signals, yet robust classification remains challenging due to inter- and intra-subject variability. EEGNet, a convolutional neural network (CNN) tailored for EEG classification, still relies on fixed temporal filters and cannot fully capture subject-specific spectral characteristics. We propose FA-EEGNet (Frequency-Adaptive Temporal Kernel EEGNet), an enhanced variant of EEGNet that replaces the fixed temporal convolution layer with a frequency-adaptive filter parameterized by spectral band power features. We evaluate EEGNet and FA-EEGNet on the open-source Clinical BCI Challenge-WCCI 2020 dataset, which comprises EEG recordings from 10 hemiparetic stroke patients performing left- and right-hand grasp motor imagery tasks. For within-subject classification, EEGNet attains a mean Cohen's kappa of 0.632 (81.6% accuracy), whereas FA-EEGNet reaches 0.912 (95.6% accuracy). For cross-subject classification with a leave-one-subject-out (LOSO) strategy, FA-EEGNet achieves a kappa of 0.767 (88.4% accuracy) versus 0.590 (79.5% accuracy) for EEGNet. These findings demonstrate that incorporating frequency adaptivity into temporal kernels substantially improves MI-EEG classification and generalization for stroke rehabilitation BCIs.

Index Terms—Brain Computer Interface (BCI), EEGNet, Motor Imagery Classification, Frequency Adaptive Classification

I. INTRODUCTION

A brain-computer interface (BCI) is a direct communication link between the brain and external devices, such as prosthetic hands, wheelchairs, or other rehabilitation systems [1] [2]. These systems record the brain's electrical activity and, through signal analysis, convert brain signals into cognitive tasks. These systems can help people with disabilities or amputations lead a better life and interact more effectively with their surroundings. Electroencephalography (EEG) is mostly used in BCIs. EEG signals are composed of rhythmic brain waves, divided into different frequency bands such as Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha or Mu (8-13 Hz), Beta (13-30 Hz), and Gamma (30 Hz <) oscillations [3]. Motor Imagery originates from the sensorimotor cortex region of the human brain, the same region involved in real-time motor movements. These alpha and beta bands undergo significant changes during motor

imagery (MI). Identifying these patterns forms the foundation for most EEG-based classification frameworks.

Over time, traditional methods have been used for classification, which depend on manual feature extraction and subject-specific tuning [4]. For this reason, Deep learning has become increasingly prevalent in this field. EEGNet has emerged as a compact CNN architecture, specifically designed for classifying EEG signals [5]. Although EEGNet eliminates the need for manual feature extraction, a fundamental limitation remained. Its temporal convolutional filters are fixed across all subjects, assuming that all individuals exhibit similar spectral characteristics during motor imagery tasks. From neurophysiological studies, variability in dominant frequencies of sensorimotor rhythms has been found, which can affect classification. To overcome this limitation, we propose an enhanced version of EEGNet, Frequency Adaptive Temporal Kernel EEGNet, which is designed to dynamically adjust to inter-subject and intra-subject variability in EEG spectral contents. This model extends the standard EEGNet0v4 framework by introducing an adaptive temporal filtering mechanism called the Frequency Adaptive Temporal Kernel (FATK), which enables the model to adjust its temporal convolutional filter based on the dominant frequency characteristics of each trial, utilizing spectral band power features. The motivation behind this model stems from the observation that fixed temporal filters may not optimally capture variability, resulting in reduced sensitivity and generalization. Our proposed FATK module can automatically tune its spectral focus, thereby reducing the need for manual parameter selection. This study includes:

- A novel architecture, Frequency-Adaptive Temporal Kernel, which dynamically adjusts the temporal filters based on the band power features of each subject.
- Comparison of the classification performance of the standard EEGNet algorithm with our proposed Frequency-Adaptive EEGNet (FA-EEGNet), and visualization and statistical analysis showing how the frequency adaptation increases the classification accuracy.

II. LITERATURE REVIEW

In the area of Brain-Computer Interface (BCI) research, the classification of motor imagery (MI) and Electroencephalo-

gram (EEG) signals is widely researched. Common Spatial Patterns (CSP) and Filter Bank Common Spatial Patterns (FBCSP) are classical feature-engineering techniques that enhance variance differences and solve subject-dependent spectral variability by dividing EEG signals into several predefined subbands, respectively [6] [7]. FBCSP remains a benchmark for MI-BCI classification, but is limited by its manual range definition [8]. Time-frequency analysis techniques (Wavelet Transform(WT) and Wavelet Packet Decomposition(WPD)) have also been applied to capture transient characteristics of non-stationary EEG signals [9] [10]. Machine-learning classifiers, such as Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM), have been used in conjunction with these methods to enhance performance on MI tasks [11] [12]. Nonetheless, handcrafted methods are sensitive to inter-session, electrode placement, and signal noise, which makes these methods less robust and less generalizable.

Deep learning shifted EEG processing towards data-driven feature extraction [13]. Early CNNs(DeepConvNet and ShallowConvNet) showed end-to-end learning possibility [14] [15]. EEGNet introduced a compact architecture for EEG classification [5], capturing both spatial and temporal patterns, making it a preferred choice for real-time MI decoding and portable BCI systems. However, these developments do not eliminate the fact that EEGNet and its variants still use fixed temporal convolutional filters [16], assuming consistent frequency properties of all subjects and trials. However, the spectral content of EEG signals of MI is widely different in individuals because of cortical reorganization, attention, fatigue, and electrode impedance variations [17]. The dominant oscillatory contents in the alpha(8-12 Hz) and beta(13-30 Hz) bands may vary across sessions or subjects, resulting in poor feature extraction [18] [19]. Adaptive methods, such as adaptive CSP, domain adaptation networks, and attention-based CNNs, have been explored but introduce computational complexity or require significant per-subject retraining [20]. Importantly, the frequency adaptivity as an architectural element of CNN-based EEG systems has rarely been explicitly included in other works [21] [22].

To fill this gap, the present work introduces Frequency-Adaptive EEGNet, an extension of EEGNet-v4 that incorporates a Frequency Adaptive Temporal Kernel (FATK). This module trains temporal convolutional filters dynamically on a per-trial basis based on spectral band power features. This model offers a high level of robustness to spectral variability, while also enhancing the compactness and interpretability of EEGNet.

III. MATERIALS AND METHODS

Our proposed pipeline (as shown in Figure 1) takes raw multi-channel EEG, preprocesses it (filtering, artifact removal, re-referencing, normalization, and epoching), and then extracts features via two parallel branches: EEGNet and a frequency-adaptive EEGNet that emphasizes key frequency bands. Their outputs are selected or combined into a compact feature vector, which is then passed to a classifier (e.g., dense layers or SVM)

to predict the target class. Finally, performance is evaluated on held-out data using metrics like accuracy and F1 to verify generalization. Fig. 1 shows the overall pipeline of the EEG preprocessing.

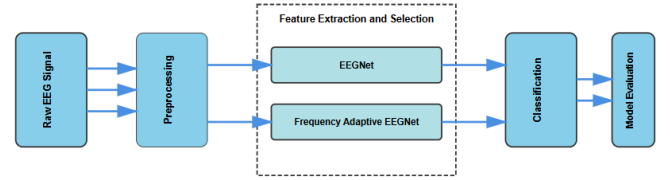


Fig. 1: Proposed EEG Processing Pipeline

A. Dataset Description

This study used an open-source EEG dataset published as part of the Clinical BCI Challenge-WCCI 2020 available at <https://github.com/Sanirban9/Clinical-Brain-Computer-Interfaces-Challenge-WCCI-2020-Glasgow>. The recordings were collected from 10 hemiparetic stroke patients showing partial paralysis or weakness of one hand after a stroke. All had either left- or right-hand motor impairments and took part in several recording sessions, which aimed to capture cortical activity related to motor attempt tasks. During data collection, participants were asked to make attempts to grasp with the left and right hands after the presentation of visual cues. Each participant's dataset consisted of 80 trials in the training file and 40 trials in the evaluation file. The trial time (including the preparatory and active motor imagery) was 8 seconds. EEG signals were collected from 12 electrodes placed following the international 10-20 system, sampled at a frequency of 512 Hz, resulting in 4096 samples per trial. Each trial has two class labels: '1' and '2' for right and left-hand grasp attempts, respectively. For noise removal, a band-pass filter(0.1-100 Hz) and a 50 Hz notch filter were applied. In this study, only the training files were used, as the evaluation files were not publicly released.

B. Preprocessing

EEG data were preprocessed using the MNE-python framework. First, the raw .mat files from the Clinical BCI dataset were imported and converted into MNE Epochs format. The trials were segmented into an interval of [-1s, 5s] relative to the cue, where negative values represent the pre-cue baseline period and positive values represent the post-cue motor imagery period. To capture the most important window, only the interval 0.5s to 4.5s was used. The initial 0.5 seconds were excluded to avoid visual processing, and the period after 4.5 seconds was discarded as it may contain anticipatory activity for the next trial. A band-pass filter(8-32 Hz) was used to remove DC offsets. The filter order was automatically determined by MNE based on the transition bandwidth to ensure proper attenuation in the stop bands. To reduce low-frequency drift, baseline correction was performed by subtracting the mean amplitude of the pre-cue interval

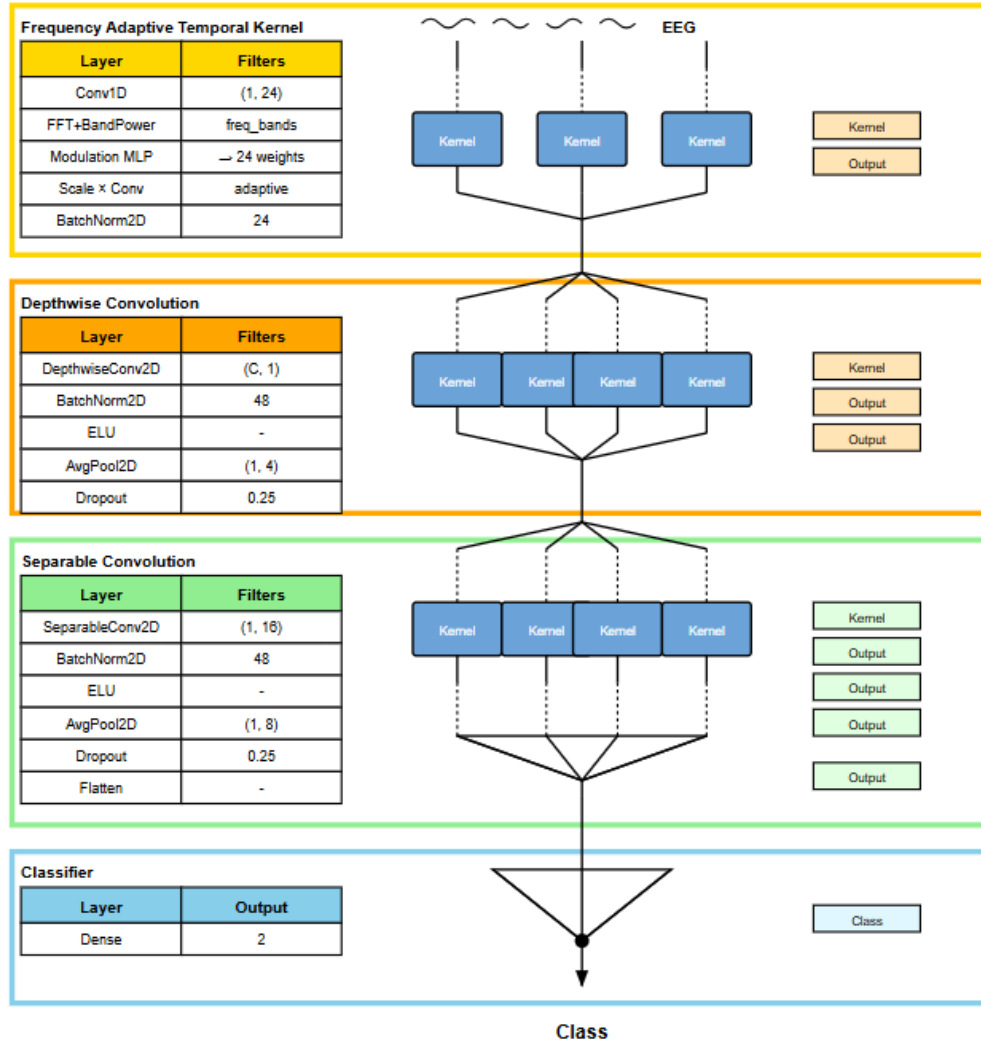


Fig. 2: Block diagram of the proposed FA-EEGNet architecture, comprising the frequency-adaptive temporal kernel module, depthwise and separable convolutional layers, and a dense classifier for motor imagery EEG-based stroke rehabilitation movement classification.

from each channel independently. Then standardization of each channel was performed to a mean of 0 and a standard deviation of 1. The preprocessed trials were arranged into a 3-dimensional shape (channels × time × trials) and passed as inputs to the deep learning model. As the dataset is small, data augmentation methods such as overlapping windows were applied, using a window size of 1024 and a 125-ms window stride. This yielded a total of 1360 samples per subject for model training.

C. Classification Model

1) *EEGNet*: EEGNet is a compact CNN model specially designed for EEG-based classification. It enables efficient learning of temporal and spatial patterns from EEG signals. The model operates through a sequence of convolutional and depthwise operations with a limited number of parameters. The

first layer acts as a band-pass filter. It learns frequency-specific features from the input. This is followed by a depthwise spatial convolution. This layer captures activation patterns across EEG channels. A separable convolution then refines temporal and spatial relationships using depthwise and pointwise operations. Finally, the classification layer converts the features into motor imagery class predictions.

2) *Proposed FA-EEGNet*: We propose a modified version of EEGNet, named Frequency-Adaptive EEGNet. Unlike the standard EEGNet, this model dynamically adjusts to subject- and trial-specific spectral variability. It employs an adaptive temporal filtering method known as the Frequency Adaptive Temporal Kernel (FATK). The frequency adaptive EEGNet follows the structure of EEGNet-v4. The network consists of four main stages: (1) Frequency-Adaptive Temporal Kernel, (2) Depthwise Spatial Convolution, (3) Separable Convolution,

and (4) Classification Head.

The FATK replaces EEGNet's fixed temporal convolution. A shared 1D convolution ($F = 24$ filters, $\text{kernel} = 128$) extracts initial features. Alpha and beta bands are computed for each trial and passed to a lightweight modulation network (two-layer MLP). This produces per-filter modulation coefficients. These coefficients scale the convolutional outputs, enabling the network to focus more on frequency elements that are most relevant to that trial. It is a form of soft gating that allows trial-specific temporal adaptation at minimal computational cost. The second layer learns spatial patterns using grouped Conv2D ($\text{kernel} = (12, 1)$, $\text{groups} = F$). Each temporal filter learns an independent spatial pattern, analogous to CSP filters. To stabilize learning and reduce redundancy, batch normalization, ELU activation, average pooling (1×4), and dropout (0.25) are used. The next layer enhances the efficiency of feature extraction by decomposing convolution into depthwise and point-wise operations (Conv2D, $\text{kernel} = (1, 16)$, $\text{groups} = 48$). It is succeeded by BatchNorm, ELU, pooling (1×8), and dropout layers. This module represents approximately 75% of the total parameters. This design allows each feature map to learn temporal dependencies independently and to be combined linearly, considerably reducing the number of parameters compared to a standard convolution layer.

The last classification Head converts the spatiotemporal features into motor imagery class predictions. The tensor output from the prior layer ($B, 48, 1, T$) is flattened into a vector of size ($B, 48 \times T$) and forwarded to a fully connected layer, which compresses it to two output neurons that correspond to the left and right hand imagery class. A softmax layer produces final class probabilities. This stage contains 2050 parameters (9.5% of the model). Fig. 2 illustrates the complete architecture of our proposed model.

D. Training Protocol

The raw EEG data are first preprocessed using standard filtering and segmentation techniques, as mentioned before. The processed signals are then passed through both the classification models. Each model outputs a class corresponding to the left and right-hand grasp movements. Model performance was primarily evaluated using Cohen's Kappa Score [23].

As the dataset contains balanced left and right-hand trials, we report both kappa and accuracy values for clarity. For the within-subject evaluation, both models used 10-fold stratified cross-validation. For each subject, the 80 training trials were divided into 10 folds. In each iteration, 9 folds were used for training and 1 fold for validation. The model was trained from scratch in each fold to avoid information leakage. The validation accuracy from the best-performing epoch was recorded for each fold. For cross-subject classification, a leave-one-subject-out (LOSO) approach was employed to facilitate better generalization to unseen subjects. In each iteration, data from 9 subjects were used for training, and the remaining subject was used for testing. Model parameters were optimized using the Adam optimizer (learning rate = 0.001) with cross-entropy loss and early stopping (patience = 15). A batch

size of 32 and a maximum of 25 epochs were used for each subject. The learning rate was adjusted using Cosine Annealing scheduler ($T_{\max} = 49$, $\eta_{\min} = 1 \times 10^{-6}$) to improve convergence stability. The model architecture consists of 12 EEG channels sampled at a rate of 512 Hz. A dropout rate of 0.5 was applied to reduce overfitting. The dropout rates and other hyperparameters were selected based on preliminary experiments.

All experiments were implemented using the Python programming language. MNE-Python and Scikit-learn packages were used for EEG preprocessing and machine learning algorithms [24] [25]. The baseline model, EEGNet, was adapted from open-source repositories to ensure consistent evaluation [17].

IV. RESULTS AND DISCUSSION

A. Training performance

The within-subject protocol showed high validation accuracy of 95.6%. The model exhibited instability during the initial training stages, with the loss rate peaking at 9.5 before stabilizing to less than 0.2 after training at epoch 15. The cross-subject model had a mean validation accuracy of 88.4. The training and validation losses steadily decreased with a total of 25 epochs of training and validation, respectively, from 0.68 to 0.28. Both approaches showed consistent late-stage convergence, confirming the advantage of subject-specific adaptation over generalized cross-subject models. Fig. 3 shows Training vs. validation loss and accuracy curves for EEG classification.

B. Confusion Matrix

The confusion matrix in Fig. 4a presents the within-subject classification performance of the FA-EEGNet model. Out of all samples, 1272 left and 1257 right-hand trials were correctly classified (TP and TN), while 88 left and 103 right-hand trials were misclassified (FP and FN). Similarly, Fig. 4b shows the cross-subject classification results: among 24,480 samples, 10,793 left and 10,839 right-hand trials were correctly classified, with 1,447 left and 1,401 right-hand trials misclassified.

C. Classification Results

The performance of the standard EEGNet and the proposed Frequency-Adaptive EEGNet is compared in both classification scenarios [26]. The Cohen's Kappa scores for all 10 subjects are summarized in Tables I and II for each of the above evaluation conditions. The within-subject classification results are summarized in Table I. EEGNet showed very consistent performance across subjects with a best (κ) of 0.632 corresponding to an accuracy of approximately 81.6%, where FA-EEGNet outperformed EEGNet for almost all subjects with an average (κ) of 0.912 corresponding to 95.6%. This improvement highlights the advantage of introducing frequency adaptivity into the model by adjusting the temporal filters.

The cross-subject results, shown in Table II, further highlight the robustness of the proposed model. In this condition, models were trained on all but one subject and tested on the unseen participant. EEGNet achieved an average (κ) of 0.590, corresponding to 79.5%, where FA-EEGNet achieved

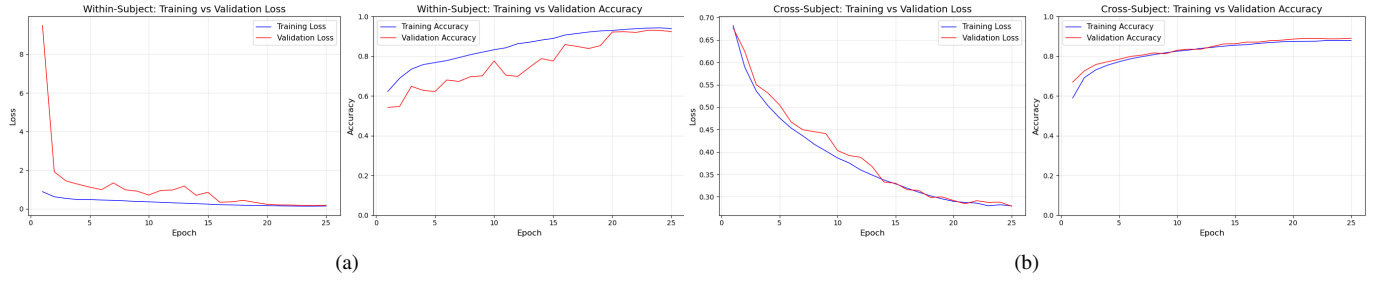


Fig. 3: Training vs. validation loss and accuracy curves for EEG classification. (a) Within subject classification. (b) Cross-subject classification.

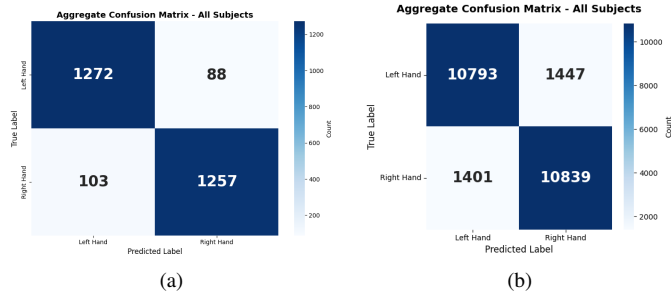


Fig. 4: Confusion matrices for motor imagery classification.(a) Within subject classification,(b)Cross-subject classification.

a significantly higher average (κ) of 0.767, corresponding to 88.4% accuracy. It consistently improved through subjects and further established that the model is capable of generalizing to unseen EEG data.

TABLE I: Within Subject Classification Results. The summary statistics are provided in the last two rows.

Subject ID	Classification Models	
	EEGNet	Proposed FA-EEGNet
P01	0.831	0.949
P02	0.971	1.000
P03	0.588	0.897
P04	0.015	0.809
P05	0.125	0.912
P06	0.684	0.934
P07	0.772	0.919
P08	0.875	0.875
P09	0.890	0.868
P10	0.566	0.956
Mean Kappa	0.632 $\pm$ 0.308	0.912 $\pm$ 0.051
Mean Accuracy	81.6 $\pm$ 0.154	95.6 $\pm$ 0.025

The substantial improvement in high-variability subjects P04 (κ : 0.015 to 0.809) and P05 (κ : 0.125 to 0.912) demonstrates FATK's ability to adapt to inconsistent spectral characteristics, while consistent cross-subject gains (mean κ : 0.590

to 0.767) indicate the model learns generalizable adaptation mechanisms rather than subject-specific overfitting.

TABLE II: Cross Subject Classification Results. The summary statistics are provided in the last two rows.

Subject ID	Classification Models	
	EEGNet	Frequency Adaptive EEGNet
P01	0.607	0.745
P02	0.591	0.780
P03	0.555	0.775
P04	0.624	0.784
P05	0.631	0.769
P06	0.574	0.737
P07	0.410	0.774
P08	0.634	0.754
P09	0.705	0.763
P10	0.574	0.792
Mean Kappa	0.590 $\pm$ 0.072	0.767 $\pm$ 0.017
Mean Accuracy	79.5 $\pm$ 0.036	88.4 $\pm$ 0.008

D. Frequency Adaptation Mechanism Analysis

Fig. 5 illustrates the adaptive behavior of FATK across different subjects and trials. The modulation heatmap shows subject-specific frequency preferences. High-alpha subjects (P02, P10) activate lower-frequency filters, while high-beta subjects (P04, P07) emphasize higher frequencies. P02 and P10 emphasize alpha (7-11Hz) while P04 and P07 prioritize beta (13-22Hz), explaining why FATK outperforms fixed filters that assume uniform responses across all subjects in the 8-13Hz range.

E. Challenges in Cross-Subject Generalization

While FA-EEGNet improved cross-subject performance, a gap of +8.9% in absolute accuracy remains between within-subject performance, revealing limitations in full subject independence. Cross-subject classification faces various challenges, including skull thickness variation, cortical anatomy differences, and spectral differences, among others. The performance is also affected by various factors, including elec-

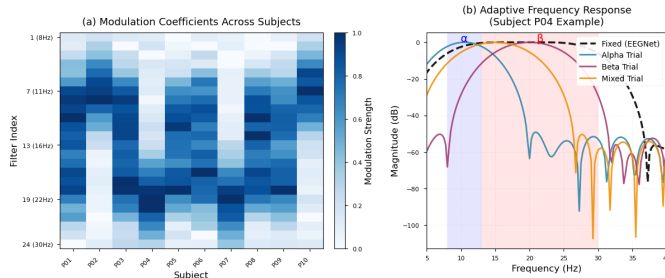


Fig. 5: Adaptive behavior of FATK across subjects and trials. (a) The modulation heatmap shows subject-specific spectral preferences. (b) Frequency responses illustrate trial-specific kernel adaptation in Subject P04 versus EEGNet’s fixed response.

trode placement, session-to-session variability, and concentration level differences, which may lead to lower accuracy compared to within-subject classification.

V. CONCLUSION

This study presents an enhanced Frequency Adaptive EEGNet (FA-EEGNet) model to improve the classification of motor imagery EEG signals, allowing for more flexible feature extraction compared to fixed filter approaches. Experimental results using the Clinical BCI Challenge-WCCI 2020 datasets show that FA-EEGNet consistently outperforms standard EEGNet across both within-subject and cross-subject classifications, achieving noticeable improvements in Cohen’s kappa and accuracy, which confirms an enhancement in generalization across subjects. Further improvements can be achieved by other adaptive mechanisms that incorporate spatial and temporal information alongside spectral information. Classification accuracy can be improved by integrating multimodal data (EMG, fNIRS) to develop a practical, subject-independent BCI framework for real-time neuroprosthetic and rehabilitation applications.

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